

Weather

Current conditions, 48-hour hourly, 5-day, animated radar. No ads, no trackers, no analytics, no accounts, no API keys. Two builds — a web PWA and a native Android app — same data, same look. Repo: [postmaster87/matts-weather-app](https://github.com/postmaster87/matts-weather-app)

Accuracy is ballpark by design — a “do I need a jacket / is that cell going to hit Veenker” app, not a life-safety product. Watches and warnings come straight from the National Weather Service and are the one thing here worth trusting.

The two builds

	Web PWA	Native Android
Where	repo root, served by GitHub Pages	android/
Built with	one HTML + one CSS + one JS file, no build step	Kotlin, Jetpack Compose, osmdroid
Install	Chrome -> Add to Home screen	sideload weather.apk
Offline	service worker + localStorage	SharedPreferences cache

What's on the screen

Section	What it shows
Alerts	Active NWS watches/warnings for the exact point. Tap one to read the full text. Hidden when there are none.
Current	Temp, condition, feels-like, today's high/low, wind + direction, gust, humidity, UV, sunrise/sunset.
Hourly	Next 48 hours, horizontal scroll. The blue bar under each hour is chance of precip.
5 Day	Day, icon, max chance of precip, and a low-to-high bar scaled across the whole 5 days. Tap a day to swap the hourly strip to that day; tap again for the rolling 48 h.
Radar	10 past frames (~1 hr) plus RainViewer's 30-minute nowcast. Opens parked on the newest observed frame — press play to run the loop. Drag the slider to scrub, Expand for fullscreen, pinch to zoom. A + on the timestamp means a forecast frame, not an observation.

Location: tap the place name to search any city, or the crosshair/pin to use GPS. The last six places stay as chips in the search screen. Fresh install defaults to Ames, IA.

Install the Android app

On your phone, in Chrome:

1. Open <https://postmaster87.github.io/matts-weather-app/weather.apk>

2. Chrome warns about the file type — accept the download.
3. Tap the downloaded file. Android asks to allow installs from Chrome; turn it on, then tap Install.
4. The launcher icon is the sun-behind-cloud, labelled **Weather**.

It is signed with a personal **release** key (4096-bit RSA, CN=Matt), not the Android debug key. Android still calls it an app “from an unknown source” — that is what sideloading looks like — but every later build signed with the same key installs straight over this one and keeps its data. No uninstall step, ever again.

The published **weather.apk** is byte-for-byte the build running on the phone:

```
sha256  cc675bf08ce52a5c52e607854938f7440d6fa0ade59a01a53f8b0ca56d153d33
```

The key is two files, neither of them in the repo, and neither recoverable:

```
%USERPROFILE%\androidkeys\matss-weather-release.p12  
android/keystore.properties          (holds the keystore password)
```

Back both up somewhere off the build machine. Lose either one and no future build can update an installed copy — the only path left is uninstall and reinstall, which loses the saved places and the cached forecast.

Install the web app instead

On your phone, in Chrome:

1. Open <https://postmaster87.github.io/matss-weather-app/>
2. Menu (three-dot) -> **Add to Home screen**

Both keep the last forecast on the device and open offline with an orange **Offline — cached 2:14pm** line at the bottom instead of pretending the numbers are current.

Rebuilding the Android app

On the Windows box — JDK 17, Android SDK 35, build-tools and Gradle 8.9 are already installed there:

```
cd android && ./gradlew assembleRelease
```

Output is android/app/build/outputs/apk/release/app-release.apk (~6.8 MB). Copy it to the repo root as weather.apk to publish it through Pages. Two files are needed for that build and neither is committed:

File	Holds
android/local.properties	sdk.dir=<path to the Android SDK>
android/keystore.properties	storeFile, storePassword, keyAlias (weather), keyPassword

Without keystore.properties the release build comes out **unsigned** rather than quietly falling back to the debug key — Gradle prints a warning saying so, and an unsigned APK refuses to install. That is a much louder failure than one signed with the wrong key, which would only surface later as an update that cannot be applied.

Data sources

Source	Used for	Key
Open-Meteo	Forecast (hourly + daily)	No
Open-Meteo Geocoding	Place search	No
api.weather.gov	NWS alerts (US only)	No
RainViewer	Radar tiles + nowcast	No
Esri dark canvas	Radar basemap	No
Leaflet / osmdroid	Map (web / Android)	No

Nothing leaves the phone except the coordinates of the place being looked up. No server, no database, no analytics.

The Android build sends `MattsWeather/1.0 (Android; +<repo URL>)` as its User-Agent, because `api.weather.gov` asks callers to identify themselves and throttles generic agents. It deliberately carries **no contact email** — add one there if you ever want NWS to be able to reach you about traffic.

Known limits

- **Radar detail caps at zoom 7.** RainViewer's free tiles do not exist past z7 — deeper zooms return a “Zoom Level Not Supported” placeholder. Both builds pin the radar layer at z7 and let the map upscale, so zooming past regional view gets blocky rather than blank. Map zoom is capped at 10.
- **Alerts are US-only.** `api.weather.gov` returns nothing outside the US; the alert strip stays hidden.
- **Radar timestamps are in the phone's timezone**, everything else is in the forecast location's local time. Only matters when looking at another time zone.
- **Daily icon vs. daily precip %** can look inconsistent (an overcast icon next to 89%). Both come from Open-Meteo as-is: the icon is the day's dominant condition, the % is the single wettest hour. Not smoothed over.
- Open-Meteo is a model blend, not a nowcast. Treat the hourly numbers as directionally right, not exact.
- **The Android app is not on Google Play** and has no update mechanism — a new version means building a new APK and installing it over the old one. Since it is signed with the release key that install is now a plain in-place update; it was an uninstall-and-lose-your-data affair while the debug key was in use.

Android source layout

MainActivity.kt	activity, permission request, theme
WeatherVm.kt	state, refresh + retry, GPS, search
data/Models.kt	Place, Current, Hour, Day, Forecast, Alert
data/WeatherApi.kt	Open-Meteo, NWS, RainViewer, geocoding
data/Net.kt	URLConnection with a real User-Agent
data/Store.kt	SharedPreferences: place, recents, cache
data/Loc.kt	LocationManager fix + platform geocoder
data/Fmt.kt	wall-clock string formatting
data/Wmo.kt	WMO code -> icon + label
ui/WeatherScreen.kt	screen, top bar, fullscreen radar
ui/Cards.kt	current / hourly / daily / alert cards
ui/RadarCard.kt	osmdroid map, frame overlays, timeline
ui/WeatherIcon.kt	the 12 condition icons, drawn on Canvas
ui/Theme.kt	palette

Icons are drawn on a Compose Canvas using the same 24-unit geometry as the web build's SVGs, so the two versions look identical rather than merely similar.